Paper Title

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Abstract

We propose an accuracy-guided method for structural channel pruning in convolutional neural networks. The method learns a binary gate for each intermediate channel by combining a classification loss with a sparsity penalty and an optional entropy term. The sparsity penalty encourages fewer active channels, while the entropy term keeps gate probabilities from becoming deterministic too early. During training, we adjust the sparsity penalty using the validation accuracy gap relative to a dense reference model. The learned gate probabilities then guide permanent channel removal, and the resulting compact network is fine-tuned. Experiments with MobileNetV2 and ResNet-50 illustrate the trade-off between model size and classification accuracy.

CCS Concepts: • **Computing methodologies** → **Neural networks**; *Supervised learning*.

Keywords: Adaptive Computation, Variational Inference, Gated Networks, Gumbel-Softmax, Layer Pruning, Dynamic Inference

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1 Introduction

Deploying convolutional neural networks under a limited parameter budget requires deciding which parts of the network can be removed while preserving predictive accuracy. Structural channel pruning addresses this problem by removing entire channels and the associated weights, yielding a narrower network. The central challenge is to determine how many channels can be removed and which ones to keep.

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Prior work approaches channel selection and conditional execution from different perspectives. AutoPruner learns binary masks for filter removal, and DepGraph identifies groups of parameters that must be pruned together to preserve architectural consistency [1, 10]. ConvNet-AIG uses binary gates to select which residual transformations to execute for each input [14]. We use binary gates during training to select intermediate channels for permanent removal. The resulting compact architecture is fixed across inputs.

Our method jointly learns the network weights and the opening probability of each channel gate. The training objective combines classification loss with a penalty on the expected number of active channels and an optional entropy term that encourages exploration during the search. When the entropy coefficient is positive, the two regularization terms admit an interpretation as a KL penalty toward a sparse Bernoulli prior. We adjust the sparsity coefficient using the validation accuracy gap to a dense reference model. Once the search is complete, the learned gate probabilities guide the construction of candidate subnetworks, and the selected subnetwork is fine-tuned after structural pruning.

Our contributions are threefold.

- We formulate channel selection with learnable Bernoulli gates and relate the entropy and sparsity terms to a sparse Bernoulli prior.
- We use validation feedback to adjust the sparsity penalty during the search, increasing it when the accuracy gap is small and relaxing it when the gap becomes too large.
- We turn learned gate probabilities into compact, fixed architectures and examine the trade-off between parameter count and test accuracy using iterative pruning on MobileNetV2 and one-shot pruning on ResNet-50.

2 Related Work

Feature selection and structured sparsification jointly learn a predictor and a smaller set of variables or computational units. Group sparse regularization groups the outgoing weights of each unit, allowing input features and hidden neurons to be selected during network training [12]. Stochastic Gates (STG) instead uses clipped Gaussian gates and a differentiable penalty on the expected number of selected input features [15]. Louizos et al. use hard-concrete gates to optimize expected L_0 regularization of network parameters [9].

These methods establish learnable gates and expected sparsity penalties as tools for selection. Our setting applies selection to intermediate convolutional channels; their physical parameter costs depend on the surrounding layers and cannot be identified with the number of selected input features or gates.

Structural channel pruning applies selection to intermediate feature maps in convolutional networks. Network Slimming uses sparsity-regularized batch normalization scaling factors [8], while AutoPruner learns binary filter masks jointly with network weights [10]. Discrimination-aware Channel Pruning alternates greedy channel selection and parameter optimization using auxiliary classification and feature-reconstruction losses [16]. CHIP evaluates channel independence through the change in the nuclear norm of feature maps when a channel is removed [13]. DepGraph identifies groups of parameters that must be pruned together to preserve architectural consistency [1].

Conditional computation changes which operations are executed for each input, without necessarily removing stored weights. ConvNet-AIG learns input-dependent binary gates for residual transformations [14]. Stochastic Depth randomly drops residual branches during training and uses the full network at test time [5]; our sampled-test baseline is an explicitly specified variant. In contrast, our channel scores are independent of the input and are converted to a fixed compact architecture. Thus active parameter counts for conditional execution and physical parameter counts after pruning describe different resources.

Herrmann et al. provide a close connection between these settings: their Gumbel-Softmax channel gates support both input-independent pruning and input-dependent computation [4]. Their batch activation penalty targets a prescribed activation rate and encourages gate polarization. Consequently, neither channel gating nor a fixed inference architecture alone distinguishes our approach. We use a penalty on mean channel-opening scores, adjust its strength from validation feedback, and distinguish threshold/budget-based iterative selection from validation-feasible one-shot checkpoint selection.

Adaptive compression also appears in DCP-Adapt, which stops channel selection based on improvement in its selection loss [16], and AMC, which uses reinforcement learning to search layer-wise compression ratios [3]. Our controller instead increases or relaxes the sparsity coefficient according to the validation accuracy gap to a dense reference. This is a feedback heuristic, not a guarantee of a prescribed final size or accuracy loss. Its effect must be assessed together with the subsequent structural selection and recovery procedure.

3 Methodology

3.1 Channel-Wise Gating

Consider a residual network with L blocks. For clarity, we describe a block with two transformations:

$$\mathbf{h}_l = a_{\theta,l}(\mathbf{x}_{l-1}), \qquad \mathbf{x}_l = s_l(\mathbf{x}_{l-1}) + b_{\theta,l}(\mathbf{h}_l), \tag{1}$$

where $a_{\theta,l}$ produces an intermediate feature map with C_l channels, $b_{\theta,l}$ maps these features to the residual output, and s_l is an identity shortcut or a projection.

We introduce one binary gate for each intermediate channel:

$$g_{l,c} \in \{0,1\}, \qquad c = 1, \dots, C_l. \tag{2}$$

The gated block becomes

$$\boxed{\mathbf{x}_l = s_l(\mathbf{x}_{l-1}) + b_{\theta,l}\left(\mathbf{g}_l \odot a_{\theta,l}(\mathbf{x}_{l-1})\right),} \tag{3}$$

where $\mathbf{g}_l = (g_{l,1}, \dots, g_{l,C_l})$ and $\odot$ denotes channel-wise multiplication broadcast over spatial dimensions.

When $g_{l,c} = 0$, intermediate channel c makes no contribution to the subsequent transformation. Because gating changes the intermediate representation rather than the block output dimensions, the residual addition remains dimensionally valid.

For a standard convolutional block, a permanently disabled intermediate channel can be removed by deleting its output filter from the preceding convolution, its associated channel-wise normalization parameters, and the corresponding input-channel slice of the following convolution.

Let

$$K = \sum_{l=1}^{L} C_l \tag{4}$$

be the total number of gated channels, and let $\mathbf{g} \in \{0,1\}^K$ collect all gate variables.

3.2 Probabilistic Formulation

For an input $\mathbf{x}$ and target $\mathbf{y}$, define

$$p_\theta(\mathbf{y}, \mathbf{g} \mid \mathbf{x}) = p_\theta(\mathbf{y} \mid \mathbf{x}, \mathbf{g}) p(\mathbf{g}), \tag{5}$$

where $p_\theta(\mathbf{y} \mid \mathbf{x}, \mathbf{g})$ is the predictive likelihood of the gated network and $p(\mathbf{g})$ is a prior over channel masks.

The marginal likelihood is

$$p_\theta(\mathbf{y} \mid \mathbf{x}) = \sum_{\mathbf{g} \in \{0,1\}^K} p_\theta(\mathbf{y} \mid \mathbf{x}, \mathbf{g}) p(\mathbf{g}). \tag{6}$$

Exact evaluation requires summing over exponentially many masks. We therefore introduce a variational distribution

$$q_\phi(\mathbf{g} \mid \mathbf{x}) = \prod_{l=1}^{L} \prod_{c=1}^{C_l} \mathrm{Bern}\left(g_{l,c} \mid r_{\phi,l,c}(\mathbf{x})\right). \tag{7}$$

Here, $r_{\phi,l,c}(\mathbf{x})$ is the probability of activating channel c in block l. A lightweight gating network predicts the probabilities from features available before the gated transformation:

$$\mathbf{r}_{\phi,l}(\mathbf{x}) = \sigma\left(h_{\phi,l}(\mathbf{z}_l(\mathbf{x}))\right), \tag{8}$$

where $h_{\phi,l}$ produces C_l logits and σ is the element-wise sigmoid function.

The factorization assumes conditional independence of the gate variables given the input. It does not imply that the resulting feature channels are independent.

3.3 Evidence Lower Bound

Applying Jensen's inequality yields

$$\begin{aligned}\log p_\theta(\mathbf{y} \mid \mathbf{x}) &= \log \mathbb{E}_{q_\phi(\mathbf{g}|\mathbf{x})}\left[\frac{p_\theta(\mathbf{y} \mid \mathbf{x}, \mathbf{g}) p(\mathbf{g})}{q_\phi(\mathbf{g} \mid \mathbf{x})}\right] \\ &\geq \mathbb{E}_{q_\phi(\mathbf{g}|\mathbf{x})}\left[\log p_\theta(\mathbf{y} \mid \mathbf{x}, \mathbf{g})\right] \\ &\quad - \mathrm{KL}\left(q_\phi(\mathbf{g} \mid \mathbf{x}) \,\|\, p(\mathbf{g})\right).\end{aligned} \tag{9}$$

Thus, the evidence lower bound is

$$\begin{aligned}\mathcal{L}_{\mathrm{ELBO}}(\theta, \phi; \mathbf{x}, \mathbf{y}) &= \mathbb{E}_{q_\phi(\mathbf{g}|\mathbf{x})}\left[\log p_\theta(\mathbf{y} \mid \mathbf{x}, \mathbf{g})\right] \\ &\quad - \mathrm{KL}\left(q_\phi(\mathbf{g} \mid \mathbf{x}) \,\|\, p(\mathbf{g})\right).\end{aligned} \tag{10}$$

For classification, the negative expected log-likelihood is the expected cross-entropy loss.

3.4 Binary Gate Sampling

For brevity, write $r_{l,c} = r_{\phi,l,c}(\mathbf{x})$ and define its logit

$$a_{l,c} = \log \frac{r_{l,c}}{1 - r_{l,c}}. \tag{11}$$

Sample independent noise variables

$$\epsilon_{l,c}^{(1)}, \epsilon_{l,c}^{(0)} \sim \mathrm{Gumbel}(0, 1). \tag{12}$$

A Bernoulli gate can be sampled as

$$g_{l,c}^{\mathrm{hard}} = \mathbf{1}\left[a_{l,c} + \epsilon_{l,c}^{(1)} - \epsilon_{l,c}^{(0)} > 0\right]. \tag{13}$$

We use the continuous relaxation

$$\widetilde{g}_{l,c} = \sigma\left(\frac{a_{l,c} + \epsilon_{l,c}^{(1)} - \epsilon_{l,c}^{(0)}}{T}\right), \tag{14}$$

where $T > 0$ is the temperature.

The straight-through gate is

$$g_{l,c}^{\mathrm{ST}} = \mathrm{stopgrad}\left(g_{l,c}^{\mathrm{hard}} - \widetilde{g}_{l,c}\right) + \widetilde{g}_{l,c}. \tag{15}$$

This construction uses binary masks in the forward pass and gradients of the relaxation in the backward pass. The resulting gradient estimator is biased.

3.5 Channel Sparsity Regularization

Choose a factorized Bernoulli prior:

$$p(\mathbf{g}) = \prod_{l=1}^{L} \prod_{c=1}^{C_l} \mathrm{Bern}(g_{l,c} \mid \pi_{l,c}). \tag{16}$$

For a single gate, with posterior activation probability r and prior activation probability π,

$$\begin{aligned}&\mathrm{KL}\left(\mathrm{Bern}(r) \,\|\, \mathrm{Bern}(\pi)\right) \\ &= r \log \frac{r}{\pi} + (1 - r) \log \frac{1 - r}{1 - \pi} \\ &= -H(r) + r \log \frac{1 - \pi}{\pi} - \log(1 - \pi),\end{aligned} \tag{17}$$

where

$$H(r) = -r \log r - (1 - r) \log(1 - r) \tag{18}$$

is the Bernoulli entropy.

Motivated by this decomposition, define

$$\mathcal{R}_{\beta,\lambda}(\phi; \mathbf{x}) = -\beta \sum_{l,c} H(r_{l,c}) + \lambda \sum_{l,c} r_{l,c}, \tag{19}$$

where $\beta > 0$ and $\lambda \geq 0$.

The complete minimization objective is

$$\boxed{\begin{aligned}\mathcal{J}(\theta, \phi) = \mathbb{E}_{(\mathbf{x},\mathbf{y})\sim\mathcal{D}}\Bigg[&\mathbb{E}_{q_\phi(\mathbf{g}|\mathbf{x})}\left[-\log p_\theta(\mathbf{y} \mid \mathbf{x}, \mathbf{g})\right] \\ &- \beta \sum_{l,c} H(r_{l,c}) + \lambda \sum_{l,c} r_{l,c}\Bigg].\end{aligned}} \tag{20}$$

Minimizing $-\beta H(r)$ encourages uncertain gate probabilities and discourages premature deterministic decisions. The sparsity term penalizes the expected number of active intermediate channels:

$$\mathbb{E}_{q_\phi(\mathbf{g}|\mathbf{x})}\left[\sum_{l,c} g_{l,c}\right] = \sum_{l,c} r_{l,c}. \tag{21}$$

This penalty measures expected channel count rather than exact computational cost. Channels in different blocks may have different costs because of their spatial resolutions and convolutional dimensions.

3.6 Interpretation as a Sparse Prior

For fixed β and λ, let all prior activation probabilities equal

$$\boxed{\pi = \sigma\left(-\frac{\lambda}{\beta}\right).} \tag{22}$$

Then

$$\beta \log \frac{1 - \pi}{\pi} = \lambda. \tag{23}$$

Consequently,

$$\begin{aligned}-\beta H(r) + \lambda r &= \beta\, \mathrm{KL}\left(\mathrm{Bern}(r) \,\|\, \mathrm{Bern}(\pi)\right) \\ &\quad + \beta \log(1 - \pi).\end{aligned} \tag{24}$$

Summing over all gates gives

$$\begin{aligned}\mathcal{R}_{\beta,\lambda}(\phi; \mathbf{x}) &= \beta\, \mathrm{KL}\left(q_\phi(\mathbf{g} \mid \mathbf{x}) \,\|\, p_\pi(\mathbf{g})\right) \\ &\quad + \beta K \log(1 - \pi).\end{aligned} \tag{25}$$

The final term is constant with respect to θ and ϕ when β and λ are fixed. Therefore, the proposed objective is equivalent, up to an additive constant, to the expected classification loss plus a weighted KL divergence.

When $\beta = 1$, this is the negative ELBO up to an additive constant. For general $\beta > 0$, it is a weighted KL-regularized variational objective; it need not be the negative of a standard ELBO.

Increasing λ at fixed β decreases π, strengthening the prior preference for inactive channels.

3.7 Interaction Between Entropy and Sparsity

For a single gate, consider only its regularizer:

$$R(r) = -\beta H(r) + \lambda r. \tag{26}$$

For $0 < r < 1$,

$$\frac{dR}{dr} = \beta \log \frac{r}{1-r} + \lambda. \tag{27}$$

Setting this derivative to zero gives

$$r^* = \sigma\left(-\frac{\lambda}{\beta}\right). \tag{28}$$

Moreover,

$$\frac{d^2R}{dr^2} = \frac{\beta}{r(1-r)} > 0, \tag{29}$$

so this is the unique minimizer of the regularizer.

Thus, the ratio λ/β determines the preferred activation probability in the absence of the prediction loss. The learned gate probabilities also depend on how each channel affects predictive performance.

A small entropy coefficient alone does not guarantee deterministic gates, and a small sparsity coefficient alone does not guarantee that all channels remain active.

3.8 Adaptive Scheduling of the Sparsity Coefficient

For the one-shot procedure in Algorithm 1, let A_t^{ref} and A_t denote the validation accuracies of the dense reference and gated model at the same search epoch t. The controller averages the last w paired gaps:

$$\overline{\Delta}_t = \frac{1}{w} \sum_{i=t-w+1}^{t} (A_i^{\text{ref}} - A_i), \qquad t \geq w. \tag{30}$$

The coefficient λ_t is used to train epoch t. For the first W warmup epochs, and until a full window is available, $\lambda_{t+1} = \lambda_t$. Subsequently, after every epoch,

$$\lambda_{t+1} = \begin{cases} \min(\lambda_{\max}, \lambda_t e^{\eta}), & \overline{\Delta}_t \leq \delta_s, \\ \max(\lambda_{\min}, \lambda_t e^{-\eta}), & \overline{\Delta}_t > \delta_h, \\ \lambda_t, & \text{otherwise}, \end{cases} \tag{31}$$

where $0 \leq \delta_s < \delta_h$ are accuracy tolerances and $\eta > 0$ is the fixed log-step. Accuracies and tolerances are fractions. This feedback rule preserves positivity and, before clipping to $[\lambda_{\min}, \lambda_{\max}]$, applies a fixed relative change.

The rule does not guarantee convergence or satisfaction of the accuracy constraint at every epoch. In particular, validation noise and delayed responses to regularization can produce oscillations around the target accuracy drop.

For fixed β, updating λ also updates the corresponding prior:

$$\pi_t = \sigma\left(-\frac{\lambda_t}{\beta}\right). \tag{32}$$

Increasing λ_t makes the prior more sparse, while decreasing it makes the prior less sparse. This is an externally controlled prior schedule rather than optimization of a single fixed-prior ELBO throughout training.

3.9 Structural Channel Pruning

Input-dependent gates produce different masks for different inputs. Permanent channel removal requires selecting a fixed mask.

We estimate the average activation probability of each channel on a representative calibration set $\mathcal{D}_{\text{cal}}$:

$$\overline{r}_{l,c} = \frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{\mathbf{x} \in \mathcal{D}_{\text{cal}}} r_{\phi,l,c}(\mathbf{x}). \tag{33}$$

Channels satisfying

$$\overline{r}_{l,c} < \varepsilon_{\text{prune}} \tag{34}$$

are candidates for permanent removal, where $\varepsilon_{\text{prune}} \in (0, 1)$ is a pruning threshold.

For the intermediate-channel placement described above, structural pruning removes the corresponding output channels of $a_{\theta,l}$ and input channels of $b_{\theta,l}$, together with associated channel-wise parameters. The output dimensions of the residual branch and shortcut remain unchanged.

This operation assumes compatible channel connectivity, as in standard dense convolutions with channel-wise normalization. Grouped convolutions and operations that mix channels require additional structural constraints.

Pruning can be performed iteratively:

1. Train the gated network.
2. Estimate average channel activation probabilities.
3. Remove a selected set of low-probability channels.
4. Fine-tune and validate the reduced network.
5. Repeat while the desired accuracy–size trade-off remains acceptable.

Low average activation probability is a pruning heuristic, not a guarantee that a channel is unnecessary for every input. Predictive performance must therefore be evaluated after structural removal.

Binary masking alone does not reduce the cost of convolutions that have already been computed. Actual computational savings require physical channel removal or an execution backend that avoids inactive computations. Any efficiency evaluation must also account for the cost of the gating modules.

Algorithm overview. Algorithm 1 specifies the one-shot procedure with input-independent gates used for ResNet-50. Here T_s, T_r are the search and recovery budgets. Section 3.8 defines the controller, including warmup, the paired averaging window, the fixed log-step and clipping. The default automatic log-step is computed once and requires $\lfloor T_s/3 - W \rfloor \geq 1$ and $\lambda_{\min} \leq \lambda_0 < \lambda_{\text{target}} \leq \lambda_{\max}$. Numerical training settings are given in Section 4.2.4.

Let (θ_t, ϕ_t) denote the saved search checkpoint, with θ_t including the weights and corresponding BatchNorm statistics, and ϕ_t the gate logits. The operator $\text{MakeMask}_{\geq 1}(\phi_t)$ keeps all channels with $r > 0.5$ in each group; if none qualify, it keeps one channel with maximal r. This minimum can activate a channel closed in the original gated validation. For each second-half checkpoint, $\text{Eval}_{\text{val}}(\theta_t, m_t)$ measures validation accuracy with the fixed mask, without updating weights or BatchNorm statistics; no training or BatchNorm recalibration occurs during selection. $\text{Params}(m_t)$ counts the parameters of the architecture with the retained layer widths, excluding gates, without constructing each compact network; it is not an expected active-parameter count.

After selection, $\text{Compact}(\theta_{t^*}, m_{t^*})$ physically removes the discarded channels and gates, transferring surviving weights and their BatchNorm state to the fixed architecture g. Next, $\text{RecalibrateBN}(g)$ updates BatchNorm statistics on training data. Recovery uses a fresh optimizer and scheduler for T_r epochs and returns the checkpoint with the highest validation accuracy among those epochs.

3.10 Iterative Pruning Procedure

Algorithm 2 makes the iterative procedure above explicit. It uses the same gate probabilities and structural removal operations, while changing how channels and recovery weights are selected. At each cycle, the final search checkpoint supplies the channel scores $\bar{r}_{l,c}$. For the input-independent gates used in these experiments, $\bar{r}_{l,c} = r_{l,c}$, so no calibration average is needed. Unlike Algorithm 1, this procedure does not select the smallest validation-feasible checkpoint from the search history.

Let D contain the permanently disabled channels. Threshold selection considers remaining channels with $\bar{r}_{l,c} < \varepsilon_{\text{prune}}$; the experimental tables denote this threshold by τ. Budget-only selection considers all remaining channels. Both retain at least one channel per gate group. When a cap ρ is specified, the removal budget is $B = \rho P_{\text{probe}}$, where P_{probe} counts parameters in the full-width gated model. The estimated cost $c(l, c)$ includes the associated convolutional and normalization parameters. These costs are additive approximations: the cap is not an achieved compression ratio or a fraction of the current compact model.

For MobileNetV2, $\mathcal{I}$ reloads ImageNet weights for every phase, using the surviving channel slices for recovery and a fresh classifier. For iterative ResNet-50, every search and recovery phase starts with random weights. Thus only D passes between phases and cycles; recovery does not inherit search weights or initialize the next search. Candidates that exceed the remaining budget are skipped. The procedure stops after recovery if no new removal was accepted, or after N cycles. Validation selects recovery checkpoints, rather than vetoing individual channel removals. One-pass MobileNetV2 sets $N = 1$ and retains these selection and initialization rules.

4 Experimental Setup

Experiments include iterative channel-pruning trajectories and single-pass pruning results on both model–dataset pairs.

4.1 Datasets

We evaluate MobileNetV2 on TinyImageNet-200 [7] (90,000 training, 10,000 validation and 10,000 test images, official validation used as test) and ResNet-50 on CIFAR-10 [6] (45,000 training, 5,000 validation and 10,000 test images). Inputs are 224×224 for TinyImageNet (upscaled from 64×64) and 32×32 for CIFAR-10. Training uses random crops, horizontal flips and dataset-specific normalization. TinyImageNet evaluation resizes to 256×256 before a 224×224 center crop; CIFAR-10 evaluation only converts and normalizes tensors.

4.2 Implementation Details

Algorithms 1 and 2 differ in channel/checkpoint selection and weight initialization, as well as the number of cycles. Table 1 summarizes the phase budgets; warmup is included in search, and reference-model training is additional.

The iterative MobileNetV2 and one-shot ResNet-50 configurations described below use AdamW, batch size 128 and a separate cosine learning-rate decay to zero in each phase. Weight decay is $5 \cdot 10^{-4}$, except for one-shot ResNet gate logits, which have zero weight decay.

4.2.1 MobileNetV2: iterative pruning. The ten-cycle run uses Algorithm 2, totaling 750 search/recovery epochs. Search and recovery restart from ImageNet weights; only the removed-channel set passes between phases. Each recovery phase contributes one validation-selected model to the trajectory.

We use MobileNetV2 [11] with its standard channel widths (2,480,072 parameters), using torchvision `IMAGENET1K_V2` weights and a new 200-way classifier. Learning rates are 10^{-4} for ImageNet-initialized weights and 10^{-3} for gates and classifier. The dense Baseline is fine-tuned for 45 epochs using the same optimizer settings.

Gate placement. We gate the hidden width of all 16 inverted residual blocks with an expansion layer, after depthwise convolution, BatchNorm and ReLU6 (ReLU clipped at 6). Each hidden gate couples one expansion filter, one depthwise filter, their BatchNorm parameters and the corresponding projection input channel. Placing the gate after normalization makes masking equivalent to physical removal; placing

Algorithm 1 Accuracy-guided structural channel pruning. Default controller settings are warmup $W = 10$ epochs, averaging window $w = 3$ epochs, and initial coefficient $\lambda_0 = 10^{-3}$. The empirically chosen clipping bounds are fixed at $\lambda_{\min} = 10^{-8}$ and $\lambda_{\max} = 80$. The default automatic log-step uses the fixed target $\lambda_{\text{target}} = 10$.

Require: Network f_θ; training and validation data; dense reference accuracies $\{A_t^{\text{ref}}\}_{t=1}^{T_s}$.
Require: Search and recovery budgets T_s, T_r; accuracy tolerances δ_s, δ_h.
// Search
Initialize ϕ with gate-opening probabilities close to one; $\lambda_1 \leftarrow \lambda_0$.
$\eta \leftarrow \log(\lambda_{\text{target}}/\lambda_0)/\lfloor T_s/3 - W \rfloor$.
for $t = 1, \ldots, T_s$ **do**
  Train (θ, ϕ) for one epoch with λ_t; measure validation accuracy A_t; save (θ_t, ϕ_t).
  if $t \le W$ or $t < w$ **then**
    $\lambda_{t+1} \leftarrow \lambda_t$.
  else
    Update λ_{t+1} by Eq. (31).
  end if
end for
// Checkpoint selection
$\mathcal{T} \leftarrow \{\lfloor T_s/2 \rfloor + 1, \ldots, T_s\}$.
for $t \in \mathcal{T}$ **do**
  $m_t \leftarrow \text{MakeMask}_{\ge 1}(\phi_t)$.
  $A_t^{\text{mask}} \leftarrow \text{Eval}_{\text{val}}(\theta_t, m_t)$.
  $P_t \leftarrow \text{Params}(m_t)$.
end for
$\mathcal{F} \leftarrow \{t \in \mathcal{T} : A_t^{\text{mask}} \ge A_{T_s}^{\text{ref}} - \delta_h\}$.
if $\mathcal{F} \neq \emptyset$ **then**
  $t^* \in \arg\min_{t \in \mathcal{F}} P_t$.
else
  $t^* \in \arg\max_{t \in \mathcal{T}} A_t^{\text{mask}}$.
end if
// Recovery
$g \leftarrow \text{Compact}(\theta_{t^*}, m_{t^*})$.
$g \leftarrow \text{RecalibrateBN}(g)$; initialize a fresh optimizer and scheduler.
Fine-tune g for T_r epochs.
return Best validation checkpoint of g during recovery.

Table 1. Pruning schedules. Each phase starts a fresh optimizer and scheduler. The iterative algorithms carry only the disabled-channel set between phases. Training epochs exclude shared dense-reference or ImageNet pretraining costs.

Procedure	Cycles	Search	Recovery	Search initialization	Recovery initialization
MobileNetV2, iterative	10	30	45	ImageNet + fresh gates	Surviving ImageNet slices
MobileNetV2, one pass	1	30	45	ImageNet + fresh gates	Surviving ImageNet slices
ResNet-50, iterative	10	120	150	New random weights	New random weights
ResNet-50, one shot	1	60	90	Random weights	Selected search weights

it before BatchNorm would allow a closed channel to become nonzero. The ten blocks with identity shortcuts also have output gates after projection BatchNorm:

$$\mathbf{x}_l = \mathbf{x}_{l-1} + \mathbf{g}_l^{\text{out}} \odot b_{\theta,l}(\mathbf{g}_l \odot a_{\theta,l}(\mathbf{x}_{l-1})). \tag{35}$$

Closing an output gate removes the corresponding projection filter and BatchNorm parameters while the shortcut preserves that channel and the block output width. Hidden and output gates follow the same objective and pruning rule: 7,104 hidden gates and 792 output gates, 7,896 in total.

Each gate has an input-independent learnable logit, initialized at $r_{l,c} \approx 0.87$ with small random perturbations. Training uses straight-through sampled masks at temperature $T = 1$; evaluation uses $r_{l,c} > 0.5$. Thus the calibration average in the Methodology section is simply $\bar{r}_{l,c} = r_{l,c}$.

Algorithm 2 Iterative structural channel pruning. Search/recovery budgets are $T_s = 30$, $T_r = 45$ epochs for MobileNetV2 and $T_s = 120$, $T_r = 150$ for ResNet-50. Both iterative runs use $N = 10$ cycles; one-pass MobileNetV2 uses the same procedure with $N = 1$.

Require: Training and validation data; initialization policy $\mathcal{I}$; search/recovery budgets T_s, T_r; maximum cycles N; selection mode, threshold $\varepsilon_{\text{prune}}$ and optional cap ρ; gate and sparsity-controller settings.

$D \leftarrow \emptyset$; recovery trajectory $\mathcal{G} \leftarrow [\,]$.
for $k = 1, \ldots, N$ **do**
  // Search
  Initialize a full-width gated network from $\mathcal{I}$; permanently mask channels in D; reset gates, optimizer, scheduler and controller.
  Train for T_s epochs with the gate objective and validation-based sparsity updates specified for the experiment.
  Read $\overline{r}_{l,c}$ from the final search checkpoint.
  $Q \leftarrow \{(l,c) \notin D : \overline{r}_{l,c} < \varepsilon_{\text{prune}}\}$ in threshold mode; $Q \leftarrow \{(l,c) \notin D\}$ in budget-only mode.
  If all remaining channels of a group belong to Q, exclude one with maximal score from Q to preserve a nonempty group.
  // Structural selection
  Sort Q by increasing $\overline{r}_{l,c}$; $R \leftarrow \emptyset$; $B \leftarrow \rho P_{\text{probe}}$ if a cap is set, otherwise $B \leftarrow \infty$.
  for channel (l,c) in Q **do**
    if $c(l,c) \le B$ **then**
      $R \leftarrow R \cup \{(l,c)\}$; $B \leftarrow B - c(l,c)$.
    end if
  end for
  $D \leftarrow D \cup R$; physically remove channels in D and all gates.
  // Recovery
  Initialize the compact network anew from $\mathcal{I}$; initialize a fresh optimizer and scheduler.
  Train for T_r epochs without gates; append the best validation checkpoint to $\mathcal{G}$.
  if $R = \emptyset$ **then**
    break.
  end if
end for
return Recovery trajectory $\mathcal{G}$; its last element is the final model.

Objective and normalization. Let M be the number of gate groups and C_j the $C_j > 0$ channels in group j not permanently masked in the current search phase. Hidden and output gates form separate groups. We average within groups, then across them:

$$S(r_\phi) = \frac{1}{M} \sum_{j=1}^{M} \frac{1}{C_j} \sum_{c \in C_j} r_{j,c}. \tag{36}$$

The implemented objective is

$$\mathcal{J}_{\text{impl}} = \mathcal{L}_{\text{CE}} + \lambda S(r_\phi). \tag{37}$$

Each group has equal weight, giving coefficient $\lambda/(MC_j)$ to each channel. The reported coefficients use this normalization.

Adaptive sparsity. We initialize $\lambda_0 = 10^{-4}$ and hold it fixed for three epochs. Updates then run every epoch, clipped to $[10^{-8}, 100]$. The reference is a separate dense model trained for 30 epochs, distinct from the 45-epoch Baseline. Its accuracy at epoch t is compared with the gated model's mean accuracy over the last three epochs. We set the soft and hard tolerances, δ_s and δ_h, to 2 and 4 percentage points, respectively. We increase λ for drops of at most δ_s, decrease it for drops above δ_h and hold it otherwise. The log-space step is $\eta = \log(1.25)\, 2^k$, $k \in \{0, 1, 2\}$. During increases, k rises by one (up to 2) if mean closing probability $1 - r_{l,c}$ over all gates grows by less than 0.02 since the previous update, resets for growth in $[0.02, 0.07]$ and otherwise stays unchanged; it also resets whenever λ is not increased.

Channel selection. Algorithm 2 uses $\tau = 0.95$ and $\rho = 0.10$. Because gates start below the pruning threshold, search must raise their scores to protect them, subject to the parameter cap. Recovery evaluates the physically compact network without gates.

4.2.2 MobileNetV2: one-shot pruning.

The seven settings use the same training, gate, controller and initialization protocol as iterative MobileNetV2, with one 30-epoch search and one 45-epoch recovery phase. They vary only the selection threshold and cap: $\tau = 0.99$ with $\rho \in \{0.6, 0.5, 0.4\}$, or $\tau = 0.95$ with $\rho \in \{0.4, 0.3, 0.2, 0.1\}$. This is one cycle of

Algorithm 2; it does not use the ResNet checkpoint selector in Algorithm 1.

4.2.3 ResNet-50: iterative pruning. Each of the three trajectories uses ten cycles of Algorithm 2, with 120 search and 150 recovery epochs per cycle (2,700 epochs in total). Every search and recovery phase starts with newly initialized random weights; search weights are not inherited. The trajectories use threshold selection with $\tau = 0.7$ or $\tau = 0.1$, or budget-only selection with $\rho = 0.05$. Only the cumulative removed-channel set is carried forward. The parameter counts and test accuracies of all ten recovery models in each series are listed in Table 9.

4.2.4 ResNet-50: one-shot pruning. Algorithm 1 specifies the 60-epoch search and 90-epoch recovery. Recovery inherits the surviving search weights, recalibrates BatchNorm on 200 training batches and starts a fresh optimizer. We test the best validation recovery checkpoint with frozen weights and BatchNorm.

We use a CIFAR-adapted ResNet-50 [2] (23.547M parameters), with a 3×3 stride-1 stem and no max-pooling. The eight repeated runs share random initialization (seed 42); the separately trained dense model supplies only the validation reference. All parameter groups start at learning rate 10^{-3}; entropy regularization and label smoothing are disabled.

Gates and regularization. Each of the 16 bottlenecks has two independently gated internal widths, after the first and second convolutions' BatchNorm and ReLU: 32 groups, with no output gates. Unlike iterative MobileNetV2, both training and evaluation use deterministic masks $r_{j,c} > 0.5$, with straight-through training gradients and initial opening probability approximately 0.98. In Eq. (36), C_j is fixed at the group's original channel count and M at 32, the total number of gate groups; these denominators remain fixed throughout search.

Sparsity control and selection. The controller averages three paired, epoch-matched validation gaps. We set the soft and hard tolerances, δ_s and δ_h, to 0.025 and 0.05 percentage points, respectively. We use a ten-epoch warmup, $\lambda_0 = 10^{-3}$, bounds $[10^{-8}, 80]$ and the default automatic log-step, which evaluates to $\eta \approx 0.9210$ for this search configuration, without the iterative MobileNetV2 closure-rate boost. From epochs 31–60 we select the smallest feasible physical model, with a 0.001 retention floor, equivalent here to retaining at least one channel per group; fallback maximizes validation accuracy. Candidates are scored before additional BatchNorm calibration.

Earlier S1/S2 tolerance sweeps used retention floors of 0.001/0.08. S1 feasible candidates came from epochs 1–60, with fallback from 31–60; both S2 windows were 31–60. The additional mean represents two selected repeats of the configuration above.

4.3 Baselines

For each architecture–dataset pair, we compare our method with a dense baseline, DepGraph [1] with random importance, AutoPruner [10], ConvNet-AIG [14], and Stochastic Depth [5].

MobileNetV2 comparisons. DepGraph uses 15 epochs of dense fine-tuning, followed by global pruning in 30 steps with target FLOPs speedups of 1.05, 1.1, 1.25, 1.5, and 2.0, and 30 epochs of post-pruning fine-tuning. The classifier is not pruned.

AutoPruner is applied to the expanded channels of the 16 blocks with expansion layers. Starting from a fine-tuned dense checkpoint, we train five models with different target sizes, using 20 epochs of channel selection followed by 25 epochs of post-pruning fine-tuning.

ConvNet-AIG uses input-dependent gates on the residual branches of the ten identity-shortcut blocks without removing weights. Models are trained from ImageNet weights for 45 epochs, with target-rate loss weight 2 and target rates $t \in \{0.01, 0.05, 0.1, 0.2, \ldots, 0.9\}$.

Stochastic Depth randomly drops the residual branches of the same ten blocks during training, with survival probabilities $p_l = 1 - \frac{l}{10}(1 - p_L)$. The expected number of parameters used per training step is $1{,}604{,}948.8 + 875{,}123.2p_L$. We set $p_L \in \{0, 0.109, 0.451, 0.680, 1\}$, corresponding to approximately 1.60M, 1.70M, 2.00M, 2.20M, and 2.48M parameters, respectively. Training comprises 30 epochs of dense fine-tuning followed by 15 epochs with Stochastic Depth. At test time, one Bernoulli mask is shared by all images in a minibatch and resampled for the next minibatch. Closed residual branches are skipped; surviving branch outputs are unscaled and BatchNorm statistics are frozen. Each minibatch executes its sampled subnetwork, while all 2.48M weights remain in storage.

ResNet-50 comparisons. On CIFAR-10, we report local reference runs under their original training protocols and parameter-count conventions. AutoPruner uses 150 epochs after a shared 150-epoch dense pretraining run; Stochastic Depth uses 200 epochs.

Stochastic Depth checkpoints are selected by full-network validation. At test time, an independent Bernoulli mask is sampled for each residual branch and minibatch (batch size 128), shared by all images in that minibatch, with frozen BatchNorm and unscaled surviving branches. We report each checkpoint's mean accuracy over five passes (seeds 42–46), without prediction ensembling.

We additionally compare with DepGraph under a matched budget of 150 training epochs per pruned model. For DepGraph, this includes any pretraining and subsequent training stages; our one-shot procedure uses 60 epochs of mask search and 90 epochs of recovery. Training our shared dense validation reference incurs an additional setup cost outside this

per-model budget. Matching epoch counts does not imply equal computational cost.

4.4 Evaluation Metrics

For the main experiments, we report test top-1 accuracy at validation-selected checkpoints. For the Baseline, our method, DepGraph and AutoPruner, we count parameters in the fixed network used for testing. ConvNet-AIG reports mean active counts over test images on TinyImageNet-200 and validation images on CIFAR-10. Stochastic Depth counts are expectations over minibatch masks; test inference follows the protocols described above.

5 Results and Analysis

5.1 Classification Accuracy

5.1.1 MobileNetV2 on TinyImageNet-200.

Iterative channel pruning. Figure 1 and Table 2 show the accuracy–size trade-off of all methods. The dense Baseline reaches 71.6% accuracy with 2.48M parameters.

Mild compression. The first two cycles of our method remove 9.9% and 18.7% of the parameters, with accuracy losses of 0.4 and 1.6 percentage points, respectively. At 2.015M parameters, our model reaches 70.0% accuracy, matching DepGraph at 2.185M. The closest AutoPruner model by parameter count reaches 68.9% at 1.900M.

Stochastic Depth reaches 71.8% at a reported expected active count of 2.2M under sampled test masks. ConvNet-AIG stays within 0.3 percentage points of the Baseline for $t \geq 0.7$, with test-mean active counts of 97.7–99.2% of its parameters. Both retain all 2.48M parameters in storage.

Stronger compression. Our final model reaches 65.8% accuracy at 1.584M parameters. AutoPruner is more accurate at a similar size, reaching 66.8% at 1.599M. Below 1.5M parameters in Figure 1, only AutoPruner and DepGraph provide comparison points. AutoPruner gives the strongest observed accuracy–size trade-off in this range, reaching 64.5% at 1.300M, compared with DepGraph's 59.2% at 1.423M.

Single-pass channel pruning. The seven one-shot MobileNetV2 models in Figure 1 and Table 2 span 1.262–2.235M parameters and 60.7–71.1% reported test accuracy. At nearly the same size, the one-shot model reaches 69.5% at 2.013M, compared with 70.0% at 2.015M on the existing iterative trajectory; at 1.813M it reaches 68.4%, compared with 68.3% at 1.817M. These observations do not show a consistent advantage for either procedure.

5.1.2 ResNet-50 on CIFAR-10.

One-shot pruning. Figure 2 and Table 3 show the trade-off between accuracy and model size for ResNet-50 on CIFAR-10. The dense baseline achieves 93.93% test accuracy with 23.547M parameters.

Our 4.660M-parameter model achieves 93.31% accuracy, reducing the parameter count by 80.2% with an accuracy loss of 0.62 percentage points. The nearest DepGraph model achieves 93.07% with 4.566M parameters. Our model therefore achieves 0.24 percentage points higher accuracy with 2.1% more parameters.

Averaging two selected repeats gives 10.805M parameters and 93.78% accuracy, corresponding to 54.1% fewer parameters and an accuracy gap of 0.15 percentage points from the dense baseline. The nearest DepGraph model achieves 93.51% with 11.569M parameters.

Stochastic Depth reaches up to 94.84% accuracy but retains all 23.547M parameters in storage.

Iterative channel pruning. Three channel-pruning trajectories in Figure 2 contain ten models each, ordered by cycle rather than by notebook row order. The threshold series $\tau = 0.7$ reaches 92.1% at 5.325M parameters before declining to 87.8% at 3.220M in its final model. The $\tau = 0.1$ series ends at 91.4% with 5.787M parameters. The budget-only series $\rho = 0.05$ retains 15.475M parameters with 93.2% accuracy at the end. Complete trajectories are given in Table 9.

Block pruning. Four historical block-pruning models in Figure 3(b) span 15.266–18.974M parameters and 92.80–93.27% reported accuracy. Their final and preceding recovery models have the same parameter count but different accuracies; they are separate checkpoints, not seed repetitions.

5.2 Computational Efficiency

5.3 Effect of Adaptive λ Scheduling

5.4 Iterative Pruning Analysis

MobileNetV2 on TinyImageNet-200. Our run yields one model per cycle (Table 2). Most parameters are removed in the first cycles: after three cycles it has removed 26.7% of the dense parameters. Between cycle 5 and the final model the size changes only from 67.6% to 63.8% of the dense model (accuracy from 66.6% to 65.8%). These are single-run observations.

5.5 Ablation Studies

We report three diagnostic studies on ResNet-50/CIFAR-10. Test accuracies are measured on the official test set after validation-based checkpoint selection. The comparisons are exploratory: their outcomes informed subsequent experiments. Each comparison has its own architecture and training protocol, as specified below. Figure 4 summarizes these studies.

Weight inheritance. We fix a pruned architecture with 17,321,706 parameters and compare inherited and randomly initialized weights. Both branches use the same 60-epoch search result followed by 90 epochs of recovery. Inherited weights reach 93.77% test accuracy, compared with 92.95% from random initialization, a difference of 0.82 percentage

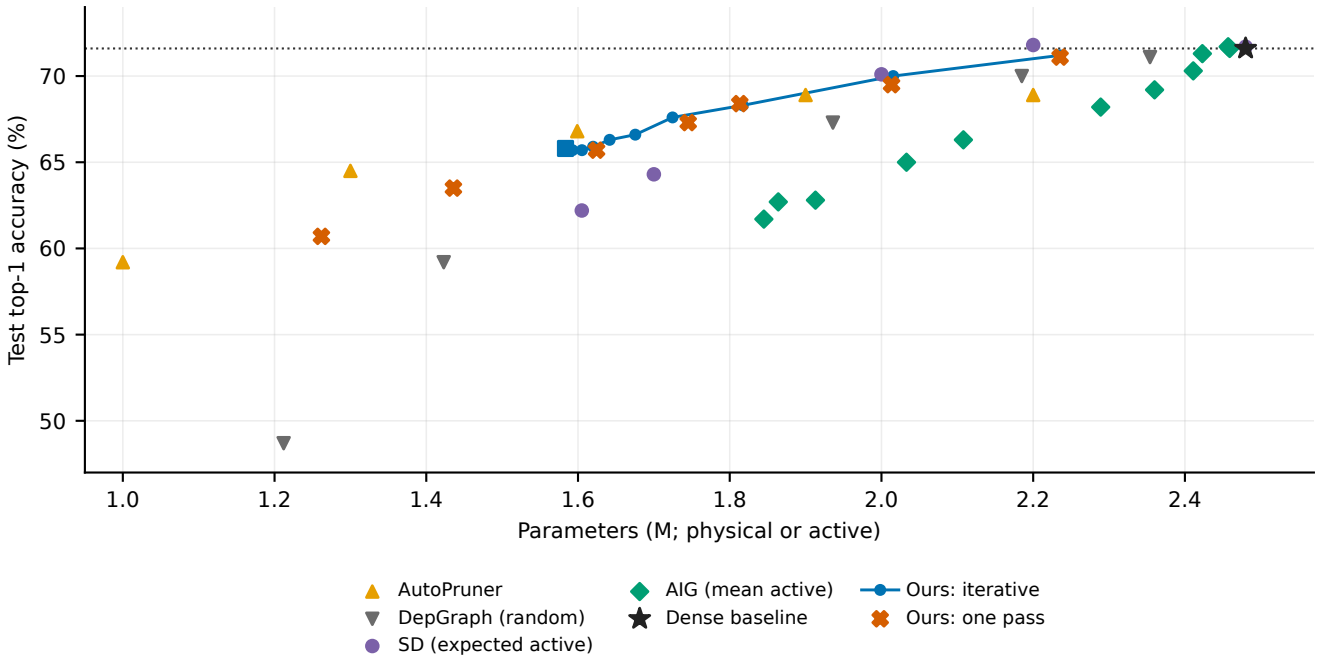


Figure 1. MobileNetV2 on TinyImageNet-200. Physical sizes of fixed networks, including all seven one-pass settings and ten iterative recovery models. Lines connect successive cycles; the square marks the final cycle and crosses denote independent one-pass settings. Dynamic methods retain all 2.48M backbone parameters; active counts are test means for AIG and expectations for SD. The dotted line is the dense baseline. Training budgets differ.

Table 2. MobileNetV2 pruning results. Physical parameter counts, share of the dense model, test accuracy and accuracy change are reported. The dense baseline has 2.480M parameters and 71.6% accuracy. Left: successive recovery models from one iterative run. Right: seven independent one-pass settings; τ is the probability threshold and ρ the configured parameter-removal cap, not the achieved parameter share. Training budgets are not matched.

Iterative pruning

Cycle	Params (M)	Share (%)	Test (%)	Δ (pp)
1	2.235	90.1	71.2	-0.4
2	2.015	81.3	70.0	-1.6
3	1.817	73.3	68.3	-3.3
4	1.725	69.5	67.6	-4.0
5	1.676	67.6	66.6	-5.0
6	1.642	66.2	66.3	-5.3
7	1.620	65.3	65.9	-5.7
8	1.605	64.7	65.7	-5.9
9	1.593	64.2	65.7	-5.9
10 (final)	1.584	63.8	65.8	-5.8

One-pass pruning

τ	ρ	Params (M)	Share (%)	Test (%)	Δ (pp)
0.99	0.6	1.262	50.9	60.7	-10.9
0.99	0.5	1.436	57.9	63.5	-8.1
0.99	0.4	1.624	65.5	65.7	-5.9
0.95	0.4	1.745	70.4	67.3	-4.3
0.95	0.3	1.813	73.1	68.4	-3.2
0.95	0.2	2.013	81.2	69.5	-2.1
0.95	0.1	2.235	90.1	71.1	-0.5

points (Table 4). This result concerns one selected architecture and one search result; it does not establish an advantage across architectures or training seeds.

Comparison with random channel selection. A separate later adaptive run reaches 93.72% test accuracy with 12.728M parameters, compared with 93.36% at 12.984M for a historical random-selection control (Table 5). Both models use 150 training epochs, but the adaptive run uses a single 60-epoch search followed by 90 epochs of recovery, whereas the random control uses an earlier iterative protocol with fixed λ. The random model also has fewer MACs: 0.692G versus 1.108G. The 0.36-point accuracy difference therefore compares complete training recipes and does not isolate channel ranking or adaptive sparsity scheduling.

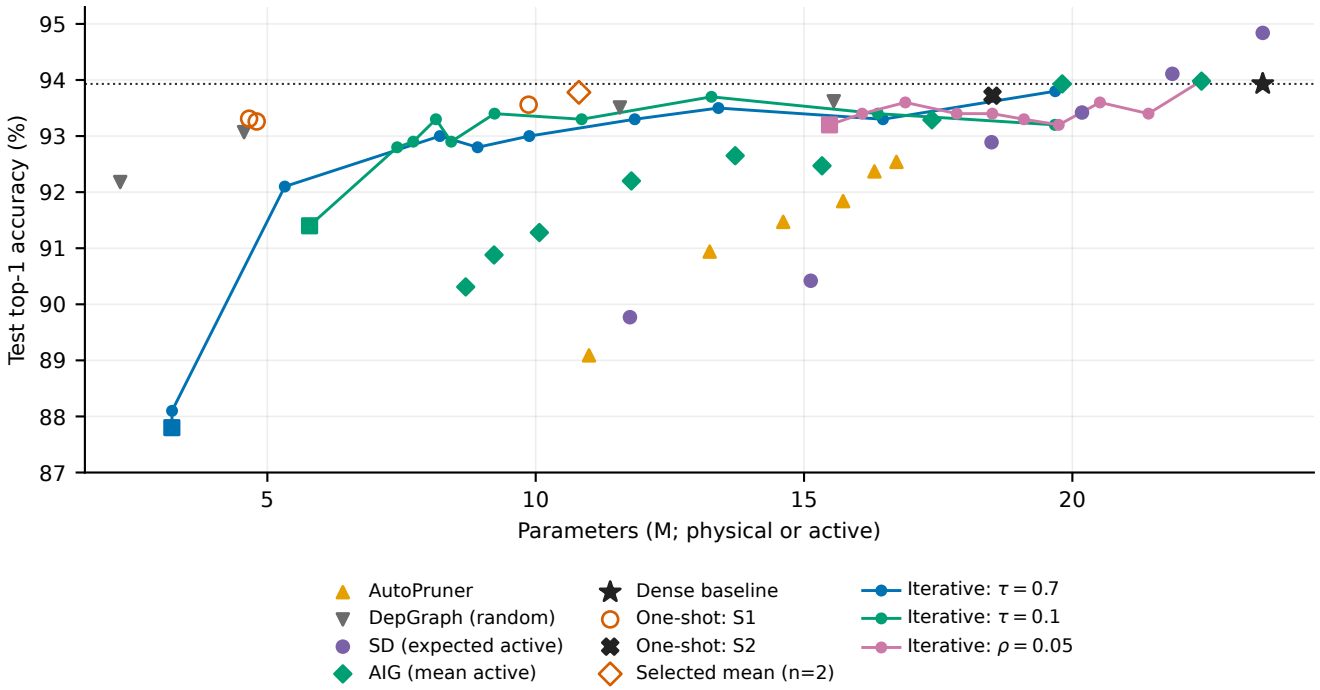


Figure 2. ResNet-50 on CIFAR-10. Fixed networks: the existing one-shot results, all 30 iterative channel-pruning models, AutoPruner and DepGraph (random importance). Lines follow cycle order and squares mark final iterative models. The hollow diamond is a coordinate-wise mean of two selected same-seed runs out of eight; it is not a deployed network. Dynamic baselines retain the full backbone: AIG counts are validation means and SD counts are expectations. Training budgets and historical accuracy aggregation differ.

Internal and output channel gates. An earlier experiment uses fixed $\lambda = 0.001$ and 150 training epochs per model to compare gates on residual-branch outputs with gates on internal bottleneck channels. Internal gating produces 12.983M parameters at 93.23% test accuracy, compared with 20.189M at 93.16% for output gating (Table 6). These runs demonstrate a smaller model at similar accuracy, but their pruning budgets and resulting architectures differ. They do not isolate gate placement at a fixed size or evaluate adaptive scheduling.

6 Discussion

6.1 Limitations

6.2 Future Work

7 Conclusion

Conclusion text.

Acknowledgments

Acknowledgments text.

Table 3. ResNet-50/CIFAR-10 one-shot and comparator results, with the three final iterative channel-pruning models. All 30 iterative checkpoints are listed in Table 9. Params: reported parameter count (M) and its share of the dense model (23,547,338 parameters, %). Δ: accuracy difference from the Baseline in percentage points. S1 and S2 denote our two sweeps; the following pair gives δ_s/δ_h in percentage points. Mean averages both coordinates of two selected runs from eight same-configuration, same-seed repeats; it is not one deployed network. Iterative settings use threshold τ or budget fraction ρ. DepGraph settings are target FLOPs speedups; k is AutoPruner's target keep ratio, p_L the final survival probability, and t the ConvNet-AIG target rate. †: expected active parameters under stochastic inference; ‡: active parameters averaged over validation images. Both marked methods retain the full backbone weights. Training budgets and historical accuracy aggregation differ.

Setting	Params (M)	%	Acc. (%)	Δ
Baseline				
dense	23.547	100.0	93.93	0.00
Ours: one-shot (selected)				
S1: 0.025/0.05	4.660	19.8	93.31	−0.62
S1: 0.25/0.5	4.801	20.4	93.26	−0.67
S1: 5/10	9.868	41.9	93.56	−0.37
Mean ($n = 2$)	10.805	45.9	93.78	−0.15
S2: 0.25/0.5	18.512	78.6	93.72	−0.21
Ours: iterative (final cycle)				
$\tau = 0.7$	3.220	13.7	87.8	-6.13
$\tau = 0.1$	5.787	24.6	91.4	-2.53
$\rho = 0.05$	15.475	65.7	93.2	-0.73
DepGraph (random importance)				
×1.5	15.554	66.1	93.62	−0.31
×2.0	11.569	49.1	93.51	−0.42
×5.0	4.566	19.4	93.07	−0.86
×10.0	2.258	9.6	92.18	−1.75
AutoPruner				
$k = 0.3$	10.993	46.7	89.09	−4.84
$k = 0.5$	13.242	56.2	90.94	−2.99
$k = 0.6$	14.612	62.1	91.47	−2.45
$k = 0.7$	15.728	66.8	91.84	−2.09
$k = 0.8$	16.312	69.3	92.37	−1.55
$k = 0.9$	16.722	71.0	92.54	−1.39
Stochastic Depth				
$p_L = 0.3$	$11.756^{\dagger}$	49.9	89.77	−4.16
$p_L = 0.5$	$15.125^{\dagger}$	64.2	90.42	−3.50
$p_L = 0.7$	$18.494^{\dagger}$	78.5	92.89	−1.04
$p_L = 0.8$	$20.179^{\dagger}$	85.7	93.42	−0.51
$p_L = 0.9$	$21.863^{\dagger}$	92.8	94.11	+0.18
$p_L = 1.0$	$23.547^{\dagger}$	100.0	94.84	+0.91
ConvNet-AIG				
$t = 0.1$	$8.696^{\ddagger}$	36.9	90.31	−3.62
$t = 0.2$	$9.224^{\ddagger}$	39.2	90.88	−3.05
$t = 0.3$	$10.068^{\ddagger}$	42.8	91.28	−2.65
$t = 0.4$	$11.783^{\ddagger}$	50.0	92.20	−1.73
$t = 0.5$	$13.717^{\ddagger}$	58.3	92.65	−1.28
$t = 0.6$	$15.335^{\ddagger}$	65.1	92.47	−1.45
$t = 0.7$	$17.383^{\ddagger}$	73.8	93.29	−0.64
$t = 0.8$	$19.811^{\ddagger}$	84.1	93.93	0.00
$t = 0.9$	$22.406^{\ddagger}$	95.2	93.98	+0.05

Table 4. Recovery initialization for a fixed architecture. Accuracy is reported in percent.

Initialization	Validation	Test
Inherited weights	94.38	93.77
Random weights	93.82	92.95

Table 5. Historical random control and the later adaptive recipe. Sizes describe physical models; test accuracy is in percent.

Method	Params (M)	MACs (G)	Test
Random selection	12.984	0.692	93.36
Our adaptive recipe	12.728	1.108	93.72

Table 6. Gate placement in the early fixed-λ experiment. Accuracy is in percent and parameter counts are physical.

Gate location	Params (M)	Validation	Test
Branch output	20.189	93.88	93.16
Bottleneck internal	12.983	94.04	93.23

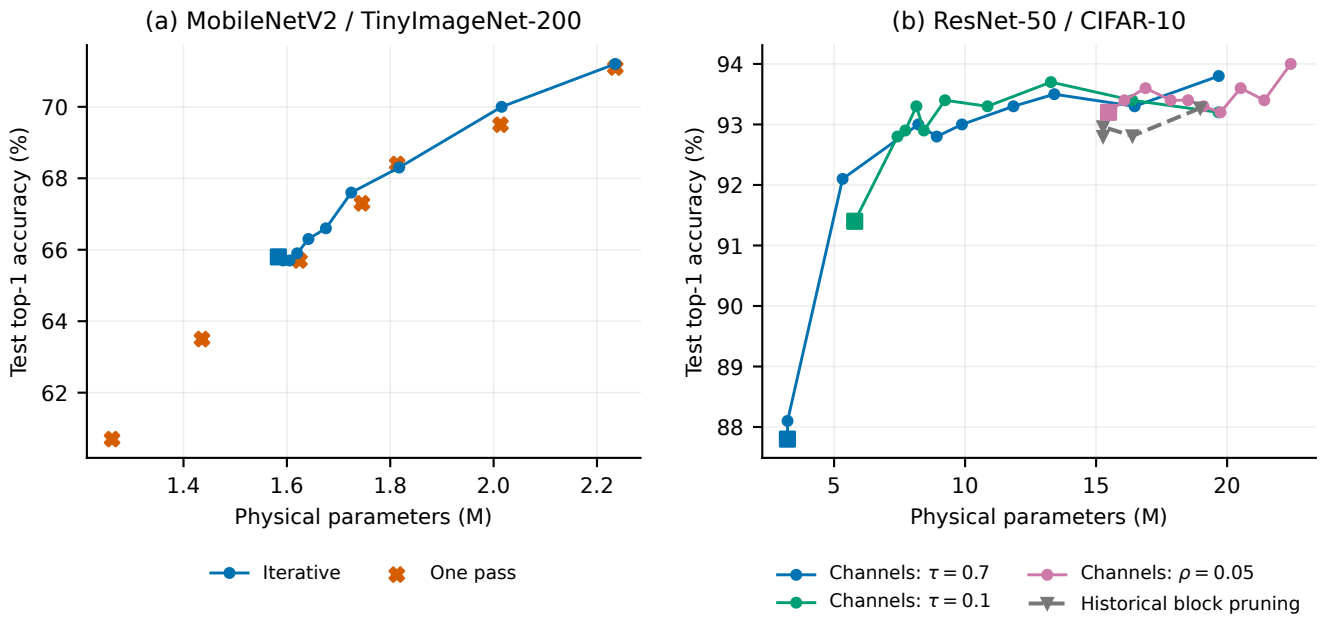


Figure 3. Pruning trajectories, isolated from the comparator sweeps. (a) Seven one-shot MobileNetV2 settings and the existing iterative trajectory. (b) Three iterative channel-pruning trajectories and one block-pruning trajectory for ResNet-50. Lines connect successive recorded cycles; the final model is placed last. Crosses in (a) are separate one-pass settings; squares mark final channel-pruning models. Counts are reported model parameters, not FLOPs or active-count estimates. No uncertainty bars or matched-budget claim are implied.

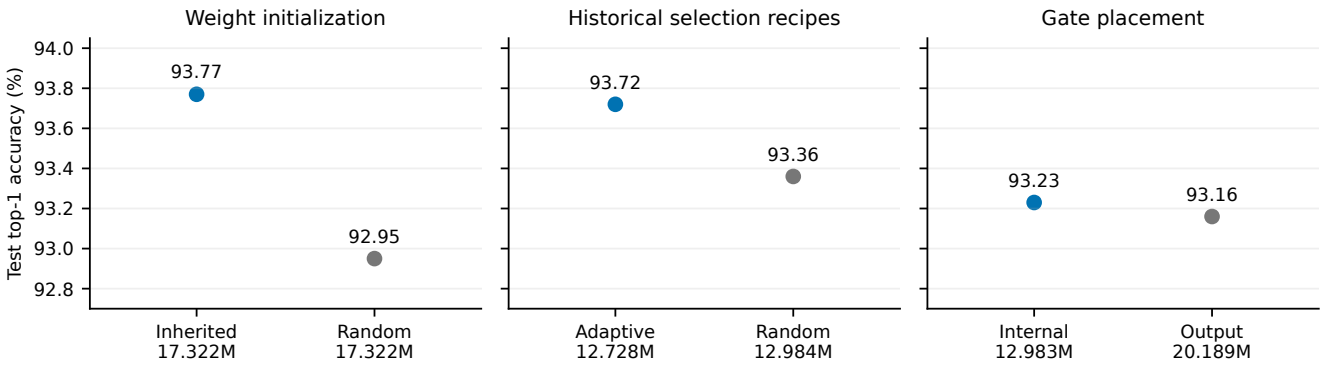


Figure 4. Three ResNet-50/CIFAR-10 diagnostics. Labels give physical parameter counts. Only the weight-initialization pair fixes the architecture and search checkpoint; selection and gate-placement comparisons use different architectures or training protocols.

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A Additional Experiments

A.1 Detailed Results

Table 7 retains all MobileNetV2 settings and results underlying Figure 1; Table 2 includes all seven additional one-pass settings.

A.2 Hyperparameter Sensitivity

Received Submission date; revised Revision date; accepted Acceptance date

Table 7. Complete MobileNetV2 results on TinyImageNet-200. Shares and accuracy changes are relative to the dense Baseline (2,480,072 parameters); pp denotes percentage points. †: test-mean active parameters for ConvNet-AIG; analytical expected active parameters for Stochastic Depth. Both store 2.48M parameters; Stochastic Depth skips closed residual branches at test time.

Baseline				
Model	**Params (M)**	**Params (% of dense)**	**Test acc. (%)**	**Acc. change (pp)**
dense	2.480	100.0	71.6	0.0
Ours				
Pruning cycle	**Params (M)**	**Params (% of dense)**	**Test acc. (%)**	**Acc. change (pp)**
1	2.235	90.1	71.2	−0.4
2	2.015	81.3	70.0	−1.6
3	1.817	73.3	68.3	−3.3
4	1.725	69.5	67.6	−4.0
5	1.676	67.6	66.6	−5.0
6	1.642	66.2	66.3	−5.3
7	1.620	65.3	65.9	−5.7
8	1.605	64.7	65.7	−5.9
9	1.593	64.2	65.7	−5.9
10 (final)	**1.584**	**63.8**	**65.8**	**−5.8**
DepGraph (random importance)				
Target FLOPs speedup	**Params (M)**	**Params (% of dense)**	**Test acc. (%)**	**Acc. change (pp)**
×1.05	2.354	94.9	71.1	−0.5
×1.1	2.185	88.1	70.0	−1.6
×1.25	1.936	78.1	67.3	−4.3
×1.5	1.423	57.4	59.2	−12.4
×2.0	1.212	48.9	48.7	−22.9
AutoPruner				
Target channel keep ratio	**Params (M)**	**Params (% of dense)**	**Test acc. (%)**	**Acc. change (pp)**
0.1809	1.000	40.3	59.2	−12.4
0.3469	1.300	52.4	64.5	−7.1
0.5125	1.599	64.5	66.8	−4.8
0.6792	1.900	76.6	68.9	−2.7
0.8452	2.200	88.7	68.9	−2.7
Stochastic Depth †				
Survival prob. p_L	**Active params (M)**	**Params (% of dense)**	**Test acc. (%)**	**Acc. change (pp)**
0.000	1.605	64.7	62.2	−9.4
0.109	1.700	68.5	64.3	−7.3
0.451	2.000	80.6	70.1	−1.5
0.680	2.200	88.7	71.8	+0.2
1.000	2.480	100.0	71.7	+0.1
ConvNet-AIG †				
Target rate t	**Active params (M)**	**Params (% of dense)**	**Test acc. (%)**	**Acc. change (pp)**
0.01	1.845	74.4	61.7	−9.9
0.05	1.864	75.1	62.7	−8.9
0.1	1.913	77.1	62.8	−8.8
0.2	2.033	82.0	65.0	−6.6
0.3	2.108	85.0	66.3	−5.3
0.4	2.289	92.3	68.2	−3.4
0.5	2.360	95.1	69.2	−2.4
0.6	2.411	97.2	70.3	−1.3
0.7	2.423	97.7	71.3	−0.3
0.8	2.457	99.1	71.7	+0.1
0.9	2.459	99.2	71.6	0.0

Table 8. Seven one-pass MobileNetV2 models from the additional notebook. τ is the probability threshold and ρ the parameter-removal cap (`thr`/`max_param` in run names); the cap is not the achieved parameter share. Each setting uses one 30-epoch search and one 45-epoch recovery phase.

τ	ρ	Params (M)	Test (%)
0.99	0.6	1.262	60.7
0.99	0.5	1.436	63.5
0.99	0.4	1.624	65.7
0.95	0.4	1.745	67.3
0.95	0.3	1.813	68.4
0.95	0.2	2.013	69.5
0.95	0.1	2.235	71.1

Table 9. Additional ResNet-50 trajectories. Channel pruning uses probability thresholds $\tau = 0.7$ and $\tau = 0.1$, or a parameter-removal budget $\rho = 0.05$ without a probability threshold; block pruning uses $\tau = 0.7$. Channel cycles are numbered 1–10 (source indices 0–8, then final). Channel accuracies are supplied to 0.1 percentage point; block values retain historical aggregation. Rows are successive recovery checkpoints, not independent repeats. The final block-pruning model has the same parameter count as its predecessor but a different checkpoint.

	Channels: $\tau = 0.7$		Channels: $\tau = 0.1$		Channels: $\rho = 0.05$	
Cycle	Params (M)	Test (%)	Params (M)	Test (%)	Params (M)	Test (%)
1	19.680	93.8	19.673	93.2	22.427	94.0
2	16.470	93.3	16.376	93.4	21.418	93.4
3	13.404	93.5	13.275	93.7	20.514	93.6
4	11.845	93.3	10.856	93.3	19.746	93.2
5	9.881	93.0	9.233	93.4	19.097	93.3
6	8.917	92.8	8.426	92.9	18.508	93.4
7	8.214	93.0	8.141	93.3	17.846	93.4
8	5.325	92.1	7.718	92.9	16.887	93.6
9	3.224	88.1	7.418	92.8	16.082	93.4
10 (final)	3.220	87.8	5.787	91.4	15.475	93.2
Block pruning: $\tau = 0.7$						
1	18.974	93.27				
2	16.384	92.81				
3	15.266	92.96				
Final	15.266	92.80				

Table 10. Earlier comparison series preserved from the additional notebook, separately from the main benchmark series. MobileNetV2 Stochastic Depth (SD): reported expected active counts; all 2.480M weights remain stored. ResNet-50 DepGraph: reported physical counts and target FLOPs speedups. DepGraph values retain only the precision of the saved notebook display. Their executed protocols are not established here.

Series / setting	Params (M)	Test (%)
MobileNetV2 / TinyImageNet-200 (SD)		
SD	1.605	64.7
SD	1.700	65.3
SD	2.000	69.0
SD	2.200	70.5
SD	2.480	70.9
ResNet-50 / CIFAR-10 (DepGraph)		
×5	4.581	92.68
×2	11.278	93.43
×10	2.247	92.07
×1.5	15.015	93.37